WORKED SOLUTION

STEP II 1989, Question 6

Pure • Integrals of Inverse Functions

STEP II 1989, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q6

The key idea

The right-hand side contains an integral of g with respect to y, so the natural move is to rewrite the left-hand integral in terms of $y = \mathrm{f}(x)$, that is, to substitute $x = \mathrm{g}(y)$. The new integrand contains $y\,\mathrm{g}'(y)$, which is crying out for integration by parts, and the boundary terms produce the rectangles. The second formula works in exactly the same way, with x replaced by $x^2/2$ and areas replaced by volumes of revolution about the y-axis.

Solution

The first formula

Since $\mathrm{f}'(x) > 0$ on $[a, b]$, f is strictly increasing there, so it is a bijection from $[a, b]$ onto $[\alpha, \beta]$ and the inverse $\mathrm{g}\colon [\alpha, \beta] \to [a, b]$ exists. Moreover g is differentiable, with

$$\mathrm{g}'(y) = \frac{1}{\mathrm{f}'(\mathrm{g}(y))},$$

which makes sense because f' never vanishes.

Substitute $x = \mathrm{g}(y)$, so that $\mathrm{f}(x) = y$ and $\mathrm{d}x = \mathrm{g}'(y)\,\mathrm{d}y$. When $x = a$ we have $y = \mathrm{f}(a) = \alpha$, and when $x = b$ we have $y = \beta$. Hence

$$\int_a^b \mathrm{f}(x)\,\mathrm{d}x = \int_\alpha^\beta y\,\mathrm{g}'(y)\,\mathrm{d}y.$$

Now integrate by parts, with $u = y$ and $\mathrm{d}v = \mathrm{g}'(y)\,\mathrm{d}y$, so $v = \mathrm{g}(y)$:

$$\begin{aligned}\int_\alpha^\beta y\,\mathrm{g}'(y)\,\mathrm{d}y &= \Big[y\,\mathrm{g}(y)\Big]_\alpha^\beta - \int_\alpha^\beta \mathrm{g}(y)\,\mathrm{d}y \\ &= \beta\,\mathrm{g}(\beta) - \alpha\,\mathrm{g}(\alpha) - \int_\alpha^\beta \mathrm{g}(y)\,\mathrm{d}y \\ &= b\beta - a\alpha - \int_\alpha^\beta \mathrm{g}(y)\,\mathrm{d}y,\end{aligned}$$

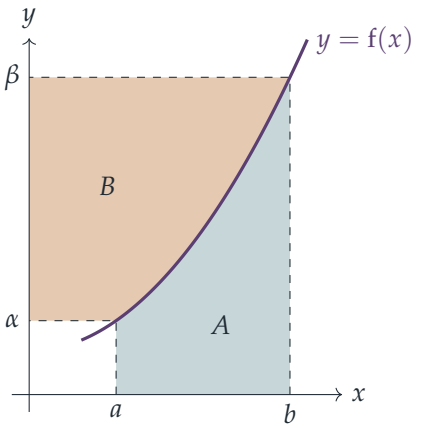
since $g(\beta) = b$ and $g(\alpha) = a$. This is the required formula.

Geometric meaning

Take $a > 0$ and $\alpha > 0$. As f is increasing, $0 < a < b$ and $0 < \alpha < \beta$, so the arc of $y = f(x)$ from (a, α) to (b, β) lies in the first quadrant. Write the formula as

$$\underbrace{\int_a^b f(x)\,dx}_{\text{area } A} + \underbrace{\int_\alpha^\beta g(y)\,dy}_{\text{area } B} = b\beta - a\alpha.$$

Here A is the area between the curve and the x-axis for $a \le x \le b$, and B is the area between the curve and the y-axis for $\alpha \le y \le \beta$ (the curve is $x = g(y)$ seen sideways). Together they fill the rectangle $[0, b] \times [0, \beta]$ with the smaller rectangle $[0, a] \times [0, \alpha]$ removed, and that L-shaped region has area $b\beta - a\alpha$.



The second formula

Use the same substitution $x = g(y)$, $dx = g'(y)\,dy$:

$$2\pi \int_a^b x\,f(x)\,dx = 2\pi \int_\alpha^\beta y\,g(y)\,g'(y)\,dy.$$

Ask yourself Which factor should we integrate this time?

Notice that $g(y)\,g'(y)$ is the derivative of $\frac{1}{2}[g(y)]^2$ by the chain rule. So integrate by parts with $u = y$ and $v = \frac{1}{2}[g(y)]^2$:

$$\begin{aligned} 2\pi \int_\alpha^\beta y\,g(y)\,g'(y)\,dy &= 2\pi \left(\left[\tfrac{1}{2} y\,[g(y)]^2\right]_\alpha^\beta - \int_\alpha^\beta \tfrac{1}{2}[g(y)]^2\,dy \right) \\ &= \pi(\beta b^2 - \alpha a^2) - \pi \int_\alpha^\beta [g(y)]^2\,dy, \end{aligned}$$

again using $g(\beta) = b$ and $g(\alpha) = a$. This is the second formula.

Geometric meaning of the second formula

Again take $a > 0$, $\alpha > 0$, and rotate the picture above through a full turn about the y-axis.

- Region A sweeps out a solid whose volume, by the method of cylindrical shells, is $2\pi \int_a^b x\,\mathrm{f}(x)\,\mathrm{d}x$: the thin strip at distance x from the axis, of height $\mathrm{f}(x)$ and width $\mathrm{d}x$, becomes a shell of volume about $2\pi x\,\mathrm{f}(x)\,\mathrm{d}x$.
- Region B sweeps out a solid whose volume, by discs, is $\pi \int_\alpha^\beta [\mathrm{g}(y)]^2\,\mathrm{d}y$: the strip at height y becomes a disc of radius $\mathrm{g}(y)$.
- Together the two solids fill a cylinder of radius b and height β with a coaxial cylinder of radius a and height α removed from its base, which has volume $\pi b^2\beta - \pi a^2\alpha$.

So the formula says shell volume plus disc volume equals the difference of the two cylinders. The conditions $a > 0$ and $\alpha > 0$ ensure that both regions lie on one side of the axis and above the x-axis, so that the integrals really are volumes and the two solids do not overlap.

Checks

Try $\mathrm{f}(x) = x^2$ on $[1, 2]$, so $\alpha = 1$, $\beta = 4$ and $\mathrm{g}(y) = \sqrt{y}$. The first formula gives $\frac{7}{3}$ on the left and $8 - 1 - \frac{2}{3}(8-1) = \frac{7}{3}$ on the right. The second gives $2\pi \cdot \frac{15}{4} = \frac{15\pi}{2}$ on the left and $\pi(16-1) - \pi \cdot \frac{15}{2} = \frac{15\pi}{2}$ on the right. We also checked both formulae numerically by Simpson's rule, with g found by bisection, for several functions including $x^3 + x$ on $[\frac{1}{2}, 2]$, where both sides of the first formula equal 5.859375.

Common slips

- Forgetting to change the limits: after substituting $x = \mathrm{g}(y)$ they become α and β, not a and b.
- Not saying why g exists and is differentiable. The condition $\mathrm{f}' > 0$ is there exactly for this.
- Describing region B as lying under a curve. It lies between the curve and the y-axis.

Beyond STEP

The first formula is the integral version of the fact that the graphs of f and f^{-1} are reflections in $y = x$. It is the idea behind Young's inequality: for increasing f with $\mathrm{f}(0) = 0$ and $a, b \geq 0$, $ab \leq \int_0^a \mathrm{f} + \int_0^b \mathrm{f}^{-1}$, and behind the Legendre transform used in convex analysis and in Hamiltonian mechanics. It is also how integrals of inverse functions such as arcsin and ln can be found without any clever substitution. For a careful treatment of substitution and inverse functions, see Abbott, *Understanding*

Analysis.

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